



SOP #13

Imaging Western Blots

***IMPORTANT:** Do NOT scratch glass inside Versadoc. No metal tools should be used around plates.

1. Turn on Black Box on benchtop (Turns on camera). You should hear a faint click.
2. Turn on Versadoc (It will go through its self-test. Lights on the door will cycle/flash).
3. Start computer program called **Quantity One**
4. Select **File**, & under tabs select **Versadoc**. The computer will then connect to Versadoc and camera
5. A **Select Application** screen should then pop up, press the select button and under **Blotting** select **chemic super sensitivity**.
6. Select **Highlight Saturation**. This will ensure if you have over-exposed the camera, the pixels will turn red. Red pixels can no longer be quantified.
7. Focus the camera :1st **red sticker on the far left** is for top plate for small western blot, 2nd red sticker is for bottom plate focus. Make sure you use **1.4** for the **aperture size** of the camera. It is the largest opening size, and will ensure all the light from your horse-radish peroxidase reaction will be recognized by the camera.
8. Wet the plate with some H₂O and place transparency containing PVDF & detection reagents on the top plate. (see instructions for preparing detection reagents below)
9. Place the membrane on the transparency, and hit **Preview** button for 1 second.
10. Press **Optimal Exposure** (stacking) button. Set exposure times (between 10/30 seconds for starting image, between 60/300 seconds for end time....determined experimentally). Press **OK** and the machine will begin the imaging process. WRITE DOWN exposure times in your book so you will remember and help guide future researchers. Exposure times can also be determined by getting image "info" either in VersaDoc or Image Lab software.

Side Note: If the Versadoc gives you an error, press **OK** and select **Optimal Exposure** again. Re-enter the exposure times and the machine should now begin working properly.

11. Take the best 2-3 photos out of the 30 you have taken.

Note: Any red areas in the photos are indicators that the area has surpassed the machines quantitation capability of the CCD (Charge Capture Device which has a limit of 56,165 photons) and the data can no longer be quantified as the area is super saturated.

12. Select an Image you think looks decent and select **Image** then **Transform** a new window should pop-up and use the **Gamma Function** to make your selected image look nicer.

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13. Select an Image you think looks decent and select **Image** then **Transform**. A new window should pop-up and use the **Gamma Function** to make your selected image look nicer.
14. Save the nice image as both a **jpeg** and a **SCN** file and put a copy on C drive, M drive and **DROP BOX**. Use the **year.month.day** format to save the image.
15. **Press print** to print a copy of the image for your notebook. **Press page setup** in print screen to set paper size to **1280 by 1280**.
16. Take Transparency out of Versadoc. Remove any excess liquid on the transparency into the green bin.
17. Use Tweezers and place PVDF back into the container with the block.
18. **Wipe down** the wet plate with kimwipe.
19. When finished with the **Versadoc turn it off** along with the **camera**, and the **computer**.

Side note: The lower shelf can be used for UV illumination (agarose gels) and trans-illumination (blue-stained protein gels) while the top shelf is used for auto-illumination like that in Western Blotting.

How to prepare PVDF for Imaging:

1. Take container containing the PVDF (membrane) and the block and bring it into Versadoc Room. Also bring tweezers and transparencies.
2. Use tweezers to remove the PVDF from block and transfers it onto a transparency.
3. Combine 1:1 Luminol and peroxide to form 850uL of ECL reagent (per sheet of PVDF).
4. Wet the PVDF using the ECL reagent ensuring area where bands will appear is completely covered in reagent.
5. Place the transparency containing the PVDF on the wet plate inside the Versadoc.